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EGR 3350-07

Technical Communications for Engineers and Computer Scientists

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Process Description

Two-Factor Authentication Verification Process

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Introduction

Two-factor authentication is a security verification process that confirms a user's identity by requiring two separate forms of authentication credentials before granting access to a system or account. The purpose of this process is to enhance protection by decreasing the chances of unauthorized entry. The process occurs in five sequential steps: entry of primary credentials, system validation and code generation, transmission of the verification code, entry and verification of the secondary code, and access authorization.

Discussion

Figure 1 illustrates the sequence of steps in the two-factor authentication process.

Entry of Primary Credentials

Entry of primary credentials is an initiation step during which a user attempts to access a secure system by entering a username and password. These aspects are submitted through a login interface where the system compares the information with stored data in an authentication database. If the credentials are valid, the process advances to the second stage; if they are incorrect, access is denied. Once the login is validated, the process moves to system validation and code generation.

System Validation and Code Generation

System validation and code generation are computational steps during which the system produces a temporary, randomly created verification code. The code remains active for a specific duration, typically between 30 and 60 seconds, to prevent the reuse of captured security codes. This serves as a secondary security measure in addition to password protection. After the code is successfully generated, it is ready for the transmission of the verification code.

Transmission of Verification Code

Transmission of verification code is a delivery process during which the system sends the generated code to the user through an alternative authentication method. The system supports methods such as text messages, emails, or mobile authentication applications. The system validates the second authentication factor by sending it to a device that the authorized user currently possesses. The user must first receive the code before they can enter and verify their secondary code.

Entry and Verification of Secondary Code

Entry and verification of the secondary code ? is a process of matching where the user enters the received code into the system for comparison. The authentication process continues when the submitted code matches the original generated value and the code remains valid. The system needs users to start a new login process when their information either becomes ~~either~~ invalid or expire~~d~~s. The final step of the process requires users to receive access authorization after they have successfully completed the matching procedure.

Access Authorization

Access authorization is a final granting step during which the system provides the user with full account access. This occurs only after both authentication factors, the password and the temporary code, have been successfully verified. This requirement of physical presence or possession of a secondary device ensures the security of the account.

Conclusion

Two-factor authentication is a security verification process that confirms a user's identity by requiring two separate forms of authentication credentials before granting access to a system or account. The process starts with the entry of primary credentials and the system generation of a code; it involves the transmission of that code to the user, followed by the entry and verification of the secondary code. Finally, the process concludes with access to authorization. By utilizing knowledge-based and possession-based methods, this process significantly decreases unauthorized access risk.

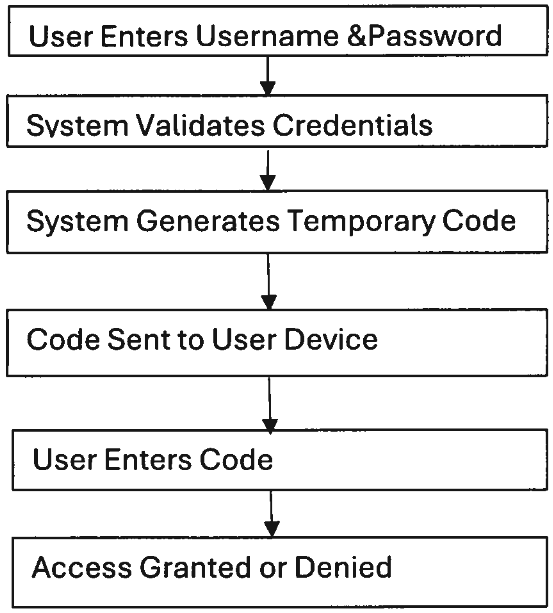

Figure 1. Two-Factor Authentication verification process. Source: Author.

Nicely done.
The last 2 steps could be combined.